

Artificial Intelligence for Corporate Finance

Session 3.5 — Valuation Prompts and Agents

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Valuation in Six prompts

Question carried forward from Session 3

How should six scarce AI interactions be allocated across a complete valuation assignment?

Message	Analytical role
1	Establish the context, supplied sources, role, limitations and decision rights.
2	Comparables Analysis under prespecified criteria.
3	Precedent Transactions Analysis under prespecified criteria.
4	DCF analysis + either reconcile now or reserve reconciliation until after review.
5	Audit the work produced so far.
6	<i>Jolly</i> : Follow-up or post-audit reconciliation or recrafting outputs.

The six prompts form one conversation (sequential).

Two defensible allocations

Prompt	Version A - reserve Prompt 6 ("the jolly")	Version B - reconciliation
1	Context, sources and decision rights	Context, sources and decision rights
2	Comparable Companies Analysis	Comparable Companies Analysis
3	Precedent Transactions Analysis	Precedent Transactions Analysis
4	DCF + valuation reconciliation	DCF
5	Mandatory audit	Mandatory audit
6	Clarify, correct, deepen, sharpen, or transform an approved output	Reconcile after the mandatory review

Version A

Preserves a genuinely flexible interaction. The first reconciliation occurs before the mandatory audit and together with DCF. Prompt 6 can address a problem that the audit identifies.

Version B

Reconciliation occurs only after the mandatory review, which gives stronger sequencing when flexibility is less valuable.

Version A's flexibility vs. Version B's structurality. No strong delegation of entire tasks to the analyst.

Prompt 1 - context, sources and decision rights

Background and context

You are an AI assistant supporting a corporate-finance analyst. You will receive a series of five prompts to complete a preliminary sell-side valuation of TargetCo for the shareholders of XXX.

Use TargetCo_company_profile.pdf, TargetCo_historical_model.xlsx, TargetCo_valuation_evidence_pack.pdf, TargetCo_management_case.pdf and TargetCo_valuation_workbook.xlsx as the supplied valuation sources. Treat SESSION3_TASK.pdf and SESSION3_DELEGATION_CANVAS.pdf as instructions, not evidence. Retain prior definitions and facts. Do not invent missing information; attribute management statements to management.

Keep explicit: i) candidate vs. selected comparable companies; ii) reported vs. management-adjusted FY2025A EBITDA; iii) Management vs. Base Case; iv) Enterprise vs. Equity Value; v) calculations vs. analyst decision.

Example: Management's EUR 1.4m adjustment is not analyst-approved merely because it was supplied.

You may extract, organize, compare and calculate. I (the analyst) retain decisions on comparison characteristics, selected companies/transactions, the historical adjustment, Base Case drivers, continuing-value assumptions and reconciliation. Stop and flag when a) information is missing, b) criteria conflict or c) judgment is required.

Later prompts will cover comparables, precedents, DCF, audit and reconciliation. Keep a concise record of assumptions, decisions, limitations and open questions. Do not begin the valuation yet. Identify only any missing or contradictory instruction that would prevent the later analysis; otherwise confirm readiness briefly.

Prompt 2 - Comparables

Analyst instruction

Using TargetCo_valuation_evidence_pack.pdf and TargetCo_valuation_workbook.xlsx, perform the Comparable Companies Analysis under the agreed rules.

Assess comparability using exactly: **[analyst-approved ranking about: sector/subsector, products/services, customers/end markets, geography..]**. Selected comparable companies: **[XXX]**. Additional company retained only for cross-checking: **[XXX]**.

Do not invent criteria, weights or scores; management's designation is part of the input, but not authority.

Produce a table (excel spreadsheet) with: candidate; characteristics specified above; management designation; missing information; conclusion under the stated criteria. Flag evidence that points in different directions. **[Add better description for the table, if needed.]** Then elucidate in a transparent and precise manner the selected companies and rationale; excluded/borderline companies and principal reason; any additional company retained only for cross-checking; EV/Revenue and EV/EBITDA trading multiples; and low, median and high for the selected set.

Separately use revenue growth, profitability, ROIC, capital intensity only to interpret differences in trading multiples. Do not use them as equal-weight selection criteria.

Apply the approved multiples to TargetCo FY2025A Revenue and separately to reported and management-adjusted FY2025A EBITDA. Label the implied Enterprise Value ranges; do not average the EBITDA bases. Cite supplied locations and list missing information and unresolved analyst decisions.

Prompt 3 - Precedent Transactions

Analyst instruction

Using TargetCo_valuation_evidence_pack.pdf and TargetCo_valuation_workbook.xlsx, perform the Precedent Transactions Analysis under the agreed rules.

Apply: **[insert the approved criteria for business comparability,]**. Selected precedent transactions: **[XXX]**.

Additional transactions retained only for comparison / cross-checking: **[XXX]**.

Do not select observations for the highest multiple or most recent date. Without an approved list, label status as proposed, and so not yet decided.

Produce a table (excel spreadsheet) with: transaction; date; business characteristics; geography; supplied buyer/transaction context; special circumstances; EV/LTM Revenue; EV/LTM EBITDA or “not available”; selected for the analysis?; rationale; principal limitation. Distinguish comparability and transaction circumstances. Identify supplied facts that could be relevant to an observed multiple. Do not claim that synergies, scarcity, bidding tension, buyer type or another narrative caused the observed transaction multiple to be higher or lower unless the supplied evidence establishes that relationship. State what was announced and what remains unestablished. Do not estimate a missing financial measure.

[Add better description for the table, if needed.]

For selected precedent transactions, show low, median and high statistics.

Apply them to TargetCo FY2025A Revenue and separately to reported and management-adjusted FY2025A EBITDA. Label Enterprise Value ranges, sources, missing information and unresolved decisions.

Prompt 4A - DCF and reconciliation

Analyst instruction — Version A

Using TargetCo_management_case.pdf and TargetCo_valuation_workbook.xlsx, there are two main tasks: i) complete the DCF under the agreed rules, then ii) reconcile it with the prior Comparable Companies Analysis and Precedent Transactions Analysis produced by the previous two prompts.

i) Regarding the DCF: Preserve the Management Case. Build the Base Case only from: **[insert each analyst-approved drivers...]**. Show revenue growth, EBITDA margin, CapEx/Revenue and operating NWC/Revenue. Calculate revenue, EBITDA, EBIT, NOPAT, D&A, CapEx, change in operating NWC and FCFF for FY2026E-FY2030E.

Forecast margins already remove the FY2025A commissioning and duplicate-quality costs: do not add the EUR 1.4m adjustment again.

Calculate continuing value under the supplied perpetuity-growth method, Management Case Enterprise Value and Base Case Enterprise Value, WACC $\times$ continuing-growth-rate sensitivity,....

Flag inconsistencies among growth, margins, reinvestment and ROIC. ii) Regarding the reconciliation, compare all three methods: method/case; Enterprise Value range; what it measures; strongest evidence; principal assumption; principal limitation; unresolved issue. Explain in a transparent way the differences in EBITDA basis, selections, forecasts and continuing-value dependence.

Propose a preliminary sell-side Enterprise Value range and corresponding Equity Value using approved net debt.

Explain how the methods support the range and identify judgments requiring approval in a precise neat manner: do not average midpoints, invent weights or collapse uncertainty.

Prompt 5 - audit

Analyst instruction

Audit every valuation output against the supplied files and recorded analyst decisions.

Recalculate totals; do not merely reread the results or the prose. Check: (1) source fidelity; (2) arithmetic/formulas; (3) reported vs. management-adjusted bases; (4) comparable-company selection against stated characteristics; (5) transaction selection against comparability....; (6) DCF formulas and signs; (7) Management vs. Base Case drivers; (8) double counting of the FY2025A adjustment; (9) Enterprise Value vs. Equity Value and net debt; (10) ... ; (11) unsupported causal claims; (12) inconsistent periods, financial measures, or reported/adjusted bases across methods; and (13) every material claim unsupported by supplied evidence. Return one table: issue; location; check; result (pass/error/unresolved); correction, if deterministic; analyst judgment required?; valuation effect.

You are allowed to correct only clear transcription, formula, or any arithmetic errors and show before/after, so to get a transparent view of the implications.

List judgment questions separately; do not change selected companies/transactions, management's adjustment, Base Case drivers, continuing assumptions or the range.

Finish with unresolved evidence gaps and the minimum synthesis before approval.

Prompt 6A - reserve the final prompt

Jolly prompt

Use the same sources, approved decisions and conversation record.

[The analyst picks **one** mode]

Mode A - follow-up. Address: [insert the audit finding, unresolved comparison, assumption/conclusion].

Do only the required following work: [clarify/.../rerun the calculation]. Show the evidence, prior to revised analysis, and valuation effect. If evidence is insufficient, stop and state what is missing. Do not reopen unrelated decisions.

Mode B - output. Use this mode only if the audit produced no correction, or after the valuation analysis/reconciliation has been updated.

Transform: [XXX] into [PDF/PowerPoint/Specified format].

Preserve the finance, sources, assumptions, ranges and unresolved issues. Improve structure and concision only.

In either mode, do not change approved calculations, selected companies or transactions, assumptions, ranges or unresolved issues unless I name the item to revise. List every change if any.

What is an agent?

What an agent has

A model + instructions

and, when needed, tools, access permissions,
rules and a required output format

What an agent does

The agent receives a task and the relevant info. It can choose the next step, use the tools available to it, observe the result and decide what to do next.

What makes it an agent? The learning loop

An agent does more than produce a single response: it can carry out a sequence of steps, choosing each next step on the basis of what it has learned so far.

The agent can act only within the instructions, tools and permissions it has been given. A multi-step task does not give it any additional authority.

One response, fixed workflow, or agent?

	Single model reply	Fixed workflow	Agent
What is decided in advance?	The input and the requested output	The sequence of steps to follow	The task, the available tools and the limits on what the agent may do
Who decides the next step?	No next step is needed	The workflow/analyst decides	The agent chooses on the basis of the information available at that point
Best suited to	A task that can be completed in one response	A task with a known and repeatable sequence	A task in which the appropriate next step depends on what happens along the way
Example	Summarize an approved report	Update the same monthly table using a predefined procedure	Explore an unusual M&A deal

Not every multi-step task requires an agent. Use an agent when choosing the next step is itself part of the task.

The agent loop: decide, act, observe, repeat

Task + available information → **Choose the next step** → **Observe the result**

Use a tool

The agent may use one of the tools available to it—for example, to retrieve information, perform a calculation or carry out an approved action.

Ask another agent

If the system includes other specialised agents, the task may be passed to one of them when appropriate.

Finish the task

If no further step is needed, the agent returns the result. If it cannot continue, it stops or asks for additional input or approval.

Why is it called a loop?

After each step, the agent uses the new information to decide what to do next. This process continues until the task is completed or the agent cannot proceed.

Define the agent before you build it

1. Objective

What should the agent accomplish?

2. What counts as complete?

What result must it produce before the task is finished?

3. Instructions and limits

What rules should it follow, and what should it avoid?

4. Information available

What data, documents or other context can it use?

5. Tools

What tools can it use to obtain information or take action?

6. Permissions

What is the agent allowed to access or change?

7. When is human approval required?

Which actions must be approved before they occur?

8. Checks and safeguards

What should be checked before an action or result is accepted?

9. When should it stop?

When should the agent stop, ask for help or report that it cannot continue?

10. How will we test it?

How will we determine whether it completed the task correctly?

Before choosing the technology

Write these decisions down first. If it is unclear what the agent should achieve, what it may do, or when it must ask for approval or stop, the design is not ready to build.

When should you use an agent?

An agent may be useful when

- length: the task requires several steps;
- dependency: what should happen next depends on the result of an earlier step;
- autonomy: the system must search for information or use tools during the task;
- unexpectancy: information may change how the task should proceed.

A simpler process is usually better when

- the required steps are known in advance;
- routine code can perform the task;
- the task is a well-defined transformation/operation on a fixed set of inputs;
- there is no useful decision for the model to make during the process.

Use an agent only when the task requires decisions that cannot be fully specified in advance.

Start with one (focused) agent

A good starting point

Begin with one agent responsible for one clearly defined task.

Consider a second agent when

Why the separation may be useful

The task is different

distinct objective and a separate definition of what counts as complete.

Different instructions

the second task must follow substantially different rules or procedures.

Different access

the tasks should not use the same tools, data or permissions.

Different approval rules

one task may require human review while another does not.

Do not add agents without a reason

Using several agents also means deciding how work is divided, how information is passed between them and how the overall result is checked. Add another agent only when the task genuinely requires that separation.

Define the task and what counts as complete

Too vague

“Monitor competitors.”

A well-defined task

Each Monday, review the approved sources for the selected competitors and identify any new information since the previous review.

Focus on changes in pricing, production capacity, acquisitions or company forecasts that meet the analyst's criteria for importance.

For each relevant change, report the source, date, what changed, why it matters and any uncertainty. Then prepare a five-bullet summary.

The task is complete when every approved source has either been reviewed or recorded as unavailable. Ask for human review when sources conflict, important information is missing or the results are intended to be shared outside the team.

Now specified

what to review; what to look for; what to produce; when the task is complete; when human review is required

Still to specify

which competitors to follow; which sources are approved; what should count as an important change

Separate general instructions from the current task

General instructions

These describe how the agent should work whenever it is used:

- what the agent is responsible for;
- which sources it may use;
- what it is allowed and not allowed to do;
- how its results should be presented;
- when human approval or review is required.

These instructions normally remain the same from one task to another.

Information for the current task

These describe the particular piece of work to be completed now:

- the period to examine;
- the files or sources relevant to this task;
- the specific question to answer;
- ... any other information that is specific to this task.

These details can change every time the agent is used.

Keep the agent's general rules separate from the info that changes from one task to another.

Give the agent only the tools it needs

Tools allow an agent to obtain information or take actions beyond producing a response.

Before giving the agent a tool, ask	What this implies
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What info does the agent need?	Give it access only to the approved sources required for the task.
What needs to change?	If the task only requires analysis, it should normally be able to read information but not modify it.
What could the tool change?	Reading a table is different from editing the table, sending a message or publishing a report.
What are the consequences of agency?	Actions such as sending, deleting, publishing or making a transaction may require additional checks or human approval.
Is the tool necessary to complete the task?	Do not give the agent tools that it does not need.
What if the tool does not work as expected?	Decide what the agent should do when information is missing, a result is unclear or the tool fails.

Give the agent the minimum set of tools needed to complete its task.

Automatic checks and human approval are different

	Automatic check	Human approval
What does it do?	Checks whether specified rules or conditions are satisfied	Asks a person to decide whether a proposed action should proceed
When is it used?	Before or after a model response, or when a tool is used	Immediately before an action that requires human authorization
Who makes the decision?	A predefined rule or automated check	An authorized person
What can happen?	The process continues, the result is rejected, or a problem is reported	The action is approved, rejected or sent back for revision

An automatic (human) check can determine whether a rule has been satisfied (authorized).

What information needs to be kept?

Information	Why it is needed	Example
Information for the current task	tells the agent what it has been asked to do now	"Review XXX against YYY"
Results obtained during the task	help the agent decide what to do next	workbook values; retrieved source.
Relevant earlier information	allows the agent to continue a task without losing important decisions or previous results	approval still required.
Official records	preserve the info that the organisation treats as authoritative	publication status

Not all information should be kept in the same place

Decide what the agent needs during the task, what must remain available if the task continues later, and what belongs in the organisation's official systems.

When should the agent stop?

Situation	What the agent should do
The task has been completed	return the required result and indicate that the task is complete.
The allowance have been exhausted	stop and report what has been completed and what remains unfinished.
A tool does not work or returns unusable information	try again only if the instructions allow it; otherwise report the problem clearly.
Important information is missing	ask for the specific information needed rather than guessing or inventing it.
A required approval is refused	do not carry out the action; revise it if requested or stop the task.
The agent cannot identify an acceptable next step	stop and explain why it cannot continue rather than acting outside its instructions.

Define both success and inability to continue

“Continue until the task is complete” is not sufficient. The agent should also know when it must stop because it cannot proceed safely or correctly.

Evaluate the process, not just the final answer

Review how the agent completed the task

For an individual task, check:

- which information and tools the agent used;
- what the tools returned;
- whether the agent followed its instructions;
- whether required checks and approvals were respected;
- whether the task was completed correctly;
- whether unnecessary or repeated steps occurred.

Test different situations

Do not test the agent only on an ordinary case. Also consider, for example:

- a required source that is missing;
- conflicting information;
- a month with no important change;
- an action that requires human approval;
- a tool that fails.

What should evaluation tell us?

A good evaluation asks two questions:

1. Did the agent produce the correct result?
2. Did it follow the correct process to get there?

Choose the technology after defining the task

Option	Useful when	What you are choosing
Agents API	the task may continue for a long time and should be able to preserve progress while it works	a more managed environment in which OpenAI handles much of the process required to continue and resume the task
Agents SDK	you are building an application with agents, tools, approvals or several specialised agents	more control over how the agent is incorporated into your own application
Responses API	you need direct model responses or want to construct the agent from simpler components	more direct control over how requests, tools and information from earlier steps are handled

Choose in this order

Define the task → **Define tools, permissions and approvals** → **Choose the technical approach** → **Build it**

Do not choose a technology first and then redesign the task around it.

What still needs to be defined?

Too vague

“Build an agent that monitors our business every month.”

Three-minute design exercise

Before deciding how to build the agent, identify the questions that still need to be answered. For each item below, write the concrete question that you would ask the business owner.

- What exactly should the agent monitor?
- Which sources is it allowed to use?
- What should count as an important change?
- What tools does it need?
- What is the agent allowed to do?
- Which actions require human approval?
- When is the task complete?
- When should the agent stop and ask for help?

Do not decide how to build the agent until these questions have clear answers.